Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The following acids have been arranged in order of decreasing acid strength. Identify the correct order ClOH(I), BrOH(II), IOH(III)

(A) I>II>III

(B) II>I>III

(C) III>II>I

(D) I>III>II

Q.2

Which of the following statements is/are wrong?

(A) Van der Waals radius of iodine is more than its covalent radius

(B) All isoelectronic ions belong to same period of the periodic table

(C) IE_1 of N is higher than that of O while IE_2 of O is higher than that of N

(D) The electron gain enthalpy of N is zero while that of P is $74.3 KJmol^{-1}$

Q.3

The alkali metal that reacts with nitrogen directly to form nitride is

(A) Li

(B) Na

(C) K

(D) Rb

Q.4

If the difference in electronegativities of two elements is very large, then

(A) The bond is 50% ionic

(B) The bond is 100% covalent

(C) The bond is more covalent than ionic

(D) The bond is more ionic than covalent

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Q.5

The correct increasing order of extent of hydrolysis is

(A) $CCl_4 < MgCl_2 < AlCl_3 < SiCl_4 < PCl_5$

(B) $CCl_4 < AlCl_3 < MgCl_2 < PCl_5 < SiCl_4$

(C) $CCl_4 < SiCl_4 < PCl_5 < AlCl_3 < MgCl_2$

(D) $CCl_4 < PCl_5 < SiCl_4 < AlCl_3 < MgCl_2$

Q.6

The correct order of melting point of chlorides of alkali metal is:

(A) $LiCl > NaCl > KCl > CsCI$

(B) $LiCl > NaCl > CsCl > KCl$

(C) $NaCl > KCl > CsCl > LiCl$

(D) $LiCl > NaCl > CsCl > KCl$

Q.7

In which of the following arrangements the order is NOT according to the property indicated against it

(A) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$: Increasing ionic size

(B) $B < C < N < O$: Increasing first ionization enthalpy

(C) $I < Br < F < Cl$: Increasing electron gain enthalpy (with negative sign)

(D) $Li < Na < K < Rb$: Increasing metallic radius

Q.8

The decreasing order of the second ionization potential of K, Ca and Ba

(A) $K > Ca > Ba$

(B) $Ca > Ba > K$

(C) $Ba > K > Ca$

(D) $K > Ba > Ca$

Q.9

Which among the following factors is the most important in making fluorine the strongest oxidizing halogen

(A) Hydration enthalpy

(B) Ionization enthalpy

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(C) Electron affinity

(D) Bond dissociation energy

Q.10

Which one of the following is the correct order for the decomposition temperature of $\mathrm{CaCO_3(X)}, \mathrm{MgCO_3(Y)}$ and $\mathrm{BaCO_3(Z)}$

(A) $Z > X > Y$

(B) $\mathrm{X > Y > Z}$

(C) $\mathrm{Y > Z > X}$

(D) $X > Z > Y$

Q.11

Which of the following is the most volatile compound?

(A) Hl

(B) HCl

(C) HBr

(D) HF

Q.12

The correct order of stability is

(A) $\mathrm{CCl_2 < SiCl_2 < SnCl_2 < PbCl_2}$

(B) $\mathrm{CCl_2 > SiCl_2 > SnCl_2 > PbCl_2}$

(C) $\mathrm{SiCl_2 > CCl_2 > SnCl_2 > PbCl_2}$

(D) $\mathrm{SnCl_2 > CCl_2 > SiCl_2 > PbCl_2}$

Q.13

The correct order of ionic radii of $\mathrm{Y^{3+}, La^{3+}}$

$\mathrm{Eu^{3+}}$ and $\mathrm{Lu^{3+}}$ is. (Atomic number, $\mathrm{Y = 39; La = 57; Eu = 63; Lu = 71}$)

(A) $\mathrm{Y^{3+} < La^{3+} < Eu^{3+} < Lu^{3+}}$

(B) $\mathrm{Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}}$

(C) $\mathrm{Lu^{3+} < Eu^{3+} < La^{3+} < Y^{3+}}$

(D) $\mathrm{La^{3+} < Eu^{3+} < Lu^{3+} < Y^{3+}}$

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Q.14

Which of the following sets of elements have almost the same atomic radii.

(A) $\mathrm{Fe, Co, Ni, Cu}$

(B) He, Ne, Ar, Xe

(C) $\mathrm{Be, Mg, Ca, Sr}$

(D) Cl, Br, I

Q.15

Melting point of calcium halides decreases in the order.

(A) $\mathrm{CaF_2 > CaCl_2 > CaBr_2 > CaI_2}$

(B) $\mathrm{CaI_2 > CaBr_2 > CaCl_2 > CaF_2}$

(C) $\mathrm{CaBr_2 > CaI_2 > CaF_2 > CaCl_2}$

(D) $\mathrm{CaCl_2 > CaBr_2 > CaI_2 > CaF_2}$

Q.16

The first ionization energy of $\mathrm{Mg, Al, P}$ and S follows the order

(A) $\mathrm{Mg < Al < P < S}$

(B) $\mathrm{Al < Mg < P < S}$

(C) $\mathrm{Al < Mg < S < P}$

(D) $\mathrm{Mg < Al < S < P}$

Q.17

Identify the correct order of acidic strength of $\mathrm{CO_2, CuO, CaO, H_2O}$

(A) $\mathrm{CaO < CuO < H_2O < CO_2}$

(B) $\mathrm{H_2O < CuO < CaO < CO_2}$

(C) $\mathrm{CaO < H_2O < CuO < CO_2}$

(D) $\mathrm{H_2O < CO_2 < CaO < CuO}$

Q.18

The atomic number 24 belongs to group 6th and 4th period. What is the atomic number of the element at the 6th period and the same group?

(A) 72

(B) 74

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(C) 56

(D) 60

Q.19

Which of the following orders are correct with respect to their size?
(1) $Al^{3+} < Mg^{2+} < Na^{+} < F^{-}$
(2) $Te^{2-} > I^{-} > Cs^{+} > Ba^{2+}$
(3) $Fe^{3+} < Fe^{2+} < Fe^{4+}$
(4) $Mg > Al > Si > P$

(A) (1), (2) & (4)

(B) (2), (3) & (4)

(C) (1), (3) & (4)

(D) (1), (2), (3) & (4)

Q.20

Which among the following elements of 3rd period shows maximum tendency of forming $d\pi - p\pi$ bond?

(A) Chlorine

(B) Silicon

(C) Phosphorus

(D) Sulphur

Q.21

Valency of silicon with respect to hydrogen is

Q.22

The element Seaborgium (Sg) has been named in the honour of Glenn T. Seaborg. What is the atomic number of that elements?

Q.23

Predict the period, group number and block of the following elements. A (at. no. $= 8$), B (at. no. $= 11), C$ (at. no. $= 28$) and D (at. no. $= 54$) are a, b, c, d respectively. Find integral value of a+2b+c+2d.

Q.24

How many of the following species are isoelectronic?

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i. Ne

ii. O^{2-}

iii. Mg^{2+}

iv. F

v. Al^{3+}

vi. Cl^{-}

vii. K

viii. Na

Q.25

Calculate the effective nuclear charge of the last electron in an atom. The electronic configuration is $1s^2, 2s^2 2p^6, 3s^2 3p^5$. (round off to the nearest integer)

Q.26

The ionisation potential of hydrogen is $13.60\mathrm{eV}$. Calculate the energy required to produce one mole of $\mathrm{H}^{\oplus}$ ion $\left(1\mathrm{eV} = 96.3\ \mathrm{kJmol}^{-1}\right)$ (round off to the nearest integer)

Q.27

The electron gain enthalpy of chlorine is $3.7\mathrm{eV}$. How much energy in kJ is released when $1\ \mathrm{g}$ of chlorine is converted completely to Cl° ion in the gaseous state. ($1\mathrm{eV} = 96.3\ \mathrm{kJ\ mol}^{-1}$) (round off to the nearest integer)

Q.28

The ionic resonance energy of $\mathrm{C-H}$ bond is $\approx 6.0\mathrm{kcalmol}^{-1}$. The EN of H is 2.1. Calculate ΔEN of carbon?
Given: $\Delta_{\mathrm{C-H}} \approx 6.0\mathrm{kcalmol}^{-1}$ (round off to the nearest integer)

Q.29

Give the atomic number of the inert gas atom in which the total number of d-electrons is equal to the difference in number to the total p - and s-electrons.

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Q.30

Calculate the electronegativity value of chlorine on Mulliken's scale, given that $\mathrm{IP} = 13.0\mathrm{eV}$ and $\mathrm{EA} = 4.0\mathrm{eV}$. (round off to the nearest integer)

Answers & Solutions

Q.1 Answer:

I>II>III

Solution:

As we move down the group basic nature of hydroxides of halogens increases so correct order of decreasing acid strength of hydroxides of halogens is I>II>III
Hence option A is correct.

Q.2 Answer:

All isoelectronic ions belong to same period of the periodic table

Solution:

In the isoelectronic series all isoelectronic anions belong to the same period and cations to the next period.

Q.3 Answer:

Li

Solution:

$6\mathrm{Li} + \mathrm{N}_2 \rightarrow 2\mathrm{Li}_3\ \mathrm{N}$ **Lithium nitride**

Q.4 Answer:

The bond is more ionic than covalent

Solution:

If the difference in electronegativities of two elements is very high then the bond is more ionic than covalent.

Q.5 Answer:

$\mathrm{CCl}_4 < \mathrm{MgCl}_2 < \mathrm{AlCl}_3 < \mathrm{SiCl}_4 < \mathrm{PCl}_5$

Solution:

Aqueous hydrolysis of CCl_4 is not possible due to absence of vacant orbital on carbon, while hydrolysis of $\mathrm{MgCl}_2, \mathrm{AlCl}_3, \mathrm{SiCl}_4$ and PCl_5 is possible. In covalent halides, hydrolysis occurs as a result of coordination of water molecules to the less electronegative element.

Q.6 Answer:

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$NaCl > KCl > CsCl > LiCl$

Solution:

Lattice energy is defined as the amount of energy required to separate one mole of solid ionic compound into its gaseous ions. Evidently greater the lattice energy, higher is the melting point of the alkali metals halide and lower is its solubility in water.

For the same halide ion, the melting point of lithium halides are lower than those of the corresponding sodium halides and thereafter they decrease as we move down the group from Na to Cs.

The low melting point of $LiCl(887\ K)$ as compared to $NaCl$ is probably because $LiCl$ is covalent in nature and $NaCl$ is ionic.

Hence the correct order is $NaCl > KCl > CsCl > LiCl$.

Q.7 Answer:

$B < C < N < O$: Increasing first ionization enthalpy

Solution:

$B < C < N < O$; When we move from B to O in a periodic table the first ionization enthalpy increases due to the attraction of the nucleus towards the outermost electron and I.E. of $N > O$.

Q.8 Answer:

$K > Ca > Ba$

Solution:

In case of alkali metals the second e^- is to be removed from a cation which has already acquired a noble gas configuration. On the other hand, in the alkaline earth metals the second e^- is to be removed from a monovalent cation (Ca^+) which still has e^- in the outer shell, thus the second e^- can be removed easily \& it becomes more easier in increasing atomic size on down to the group.

Hence the correct decreasing order of second I.P is $K > Ca > Ba$

Q.9 Answer:

Bond dissociation energy

Solution:

The bond dissociation energy of the F-F bond is very low. The weak F-F bond makes fluorine the strongest oxidising halogen.

Q.10 Answer:

$Z > X > Y$

Solution:

The decomposition temperature of carbonates of Ca, Mg and Ba follow the following order.

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$BaCO_3 > CaCO_3 > MgCO_3$
$(1633\ K)(1173\ K)(813\ K)$
$Z > X > Y$

It is due to high crystal lattice energy \& hence high stability

Q.11 Answer:

HCl

Solution:

Boiling point of HF is highest due to H-bonding. For other halogen acids b.p. increases in the order $HCl < HBr < HI$. Therefore, most volatile (with Lower b.pt) is HCl

Q.12 Answer:

$CCl_2 < SiCl_2 < SnCl_2 < PbCl_2$

Solution:

MX_2 **Stability of di halides increases, down to the group due to inert pair effect**

Q.13 Answer:

$Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$

Solution:

Correct order is $= Y^{3+} < Lu^{3+} < Eu^{3+} < La^{3+}$ **due to lanthanide contraction**

Q.14 Answer:

Fe, Co, Ni, Cu

Solution:

The atomic radii of one series of transition metals do not vary greatly because inner $3\ d$ atomic orbitals are being filled, although there is a general decrease across a series of transition elements

Q.15 Answer:

$CaF_2 > CaCl_2 > CaBr_2 > CaI_2$

Solution:

Melting point of calcium halide decreases in the order.

$CaF_2 > CaCl_2 > CaBr_2 > CaI_2$

Size of Ca^{2+} ions are comparatively smaller than halide ions. Since Ca belongs to the alkaline earth metal group the strength of the bond is higher when the two ions have comparable sizes. The size of

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Ca^{2+} is similar to fluorine ions. So stronger is the bond, while the bond becomes less strong as we move down to the group. So the melting point will also decreases

Q.16 Answer:

$Al < Mg < S < P$

Solution:

In a period, the value of ionization potential increases from left to right with breaks where the atoms have somewhat stable configurations. Due to stable configurations of Mg and P in 3 rd period have higher values than expected

Element	$P >$	$S >$	$Mg >$	Al
(IE_1)	11.0	10.4	7.6	6.0

Q.17 Answer:

$CaO < CuO < H_2O < CO_2$

Solution:

$CO_2 \longrightarrow$ acidic, $CuO \longrightarrow$ weak base

$CaO \longrightarrow$ basic, $H_2O \longrightarrow$ neutral So, correct order of acidic strength is $CaO < CuO < H_2O < CO_2$

Q.18 Answer:

74

Solution:

$24 + 18 + 32 = 74$

Q.19 Answer:

(1), (2) & (4)

Solution:

(1) Mg^{2+}, Na^{+}, Al^{3+} and F^{-} are isoelectronic and thus follow the order $_{13}Al^{3+} <_{12} Mg^{2+} <_{11} Na^{+} <_{9} F^{-}$

(2) Atomic number: $Te = 52; I = 53; Cs = 55; Ba = 56$. All the ions given are isoelectronic and have the xenon electron configuration.

Ionic radius $\propto \frac{1}{\text{nuclearcharge}}$

Hence, the given order is correct.

(3) More the positive charge on the atom, less is the size.

The correct order would be

$Fe^{+2} < Fe^{+3} < Fe^{+4}$

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Option 3 is incorrect.

(4) Across the period, the size of atoms decreases. $\mathrm{Mg} = 160\mathrm{pm}; \mathrm{Al} = 143\mathrm{pm}; \mathrm{Si} = 118; \mathrm{P} = 110\mathrm{pm}$

Q.20 Answer:

Chlorine

Solution:

Among the given elements Cl **has maximum tendency of** $\mathrm{d}\pi - \mathrm{p}\pi$ **bonding On moving across a period, nuclear charge increases and more s, p-electrons are added. Due to incomplete shielding of nuclear charge, the atomic size and so the size of 3 d-orbitals decreases from** $\mathrm{Si} < \mathrm{P} < \mathrm{S} < \mathrm{Cl}$

Decrease in size of 3d-orbitals leads to stronger $\mathrm{d}\pi$ **- p** π **overlap.**

Q.21 Answer:

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Solution:

Valence Electrons are the total number of electrons in the outermost shell.

Valency of an element is its combining capacity.

Number of valence electrons in silicon is 4 . Its compound with hydrogen is $\mathrm{SiH_4}$**. Valency of silicon is 4 .**

In case of sulphur, valence electrons are 6 . The compound formed is $\mathrm{H_2\ S}$**. Valency of sulphur is 2 .**

In case of phosphorus, valence electrons are 5. Compound is $\mathrm{PH_3}$**,** $(\mathrm{PH_5}$ **does not exist). The valency of phosphorus is 3 .**

In case of chlorine, valence electrons are 7. Compound formed is HCl**. Valency of chlorine is** 1**.**

Thus, only in case of Si, both the number of valence electrons and valency is the same with respect to hydrogen.

Q.22 Answer:

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Solution:

Seaborgium is a synthetic chemical element with the symbol Sg **and atomic number** 106 **. It is named after the American nuclear chemist Glenn T. Seaborg.**

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Q.23 Answer:

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Solution:

Electronic configurations of different elements are

$\mathrm{A} \rightarrow 1s^2 2s^2 2p^4$

$\mathrm{B} \rightarrow 1s^2 2s^2 2p^6 3s^1$

$\mathrm{C} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^8 4s^2$

$\mathrm{D} \rightarrow 1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10} 4s^2 4p^6 4d^{10} 5s^2 5p^6$

Element A: p-block element

Group number $= 10+$ **number of electron in the valence shell** $= 10 + 6 = 16$

Period of the element = Principal quantum number of the valence shell $= 2$ **nd.**

Elements B: s-block element

Group number = Number of electrons in valence shell $= 1$

Period number $= 3\mathrm{rd}$

Element C: d-block element

Group number = Number of electrons in penultimate shell and valence shell $= 8 + 2 = 10$

Period of the = Principal quantum number of the valence shell

$= 4\mathrm{th}$

Element D: p-block element

Group number $= 10+$ **number of electron in the valence shell** $= 10 + 8 = 18$

Period number $= 5$

Q.24 Answer:

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Solution:

The number of electrons in these species are:

i. $\mathrm{Ne} = 10$

ii. $\mathrm{O}^{2-} = 8 + 2 = 10$

iii. $\mathrm{Mg}^{2+} = 12 - 2 = 10$

iv. $\mathrm{F} = 9$

v. $\mathrm{Al}^{3+} = 13 - 3 = 10$

vi. $\mathrm{Cl}^{\Theta} = 17 + 1 = 18$

vii. $K = 19$

viii. $\mathrm{Na} = 11$

Thus $\mathrm{Ne}, \mathrm{O}^{2-}, \mathrm{Mg}^{2+}, \mathrm{Al}^{3+}$ **are isoelectronic species (each one of which has** 10 **electrons).**

Q.25 Answer:

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Solution:

$\mathrm{Z} = 2 + 2 + 6 + 2 + 5 = 17$**. and this atom has 3 shells in all.**

$$\therefore \sigma = \left[\begin{array}{l} 0.35 \times \text{ Number of electrons left} \\ \text{in } n\text{th shell } + (0.85 \times \text{ Number of} \\ \text{electrons in } (n-1) \text{ th shell } + (1.00 \times \\ \text{Total number of electrons in the} \\ \text{inner shells}) \end{array}\right]$$

$= (0.35 \times 6) + (0.85 \times 8) + (1 \times 2)$

$= 10.9$

Thus, $Z_{\text{eff}} = (Z - \sigma) = 17 - 10.9 = 6.1$

Q.26 Answer:

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Solution:

The ionisation potential may be represented as

$$\mathrm{H}_{(\mathrm{g})}+13.60 \mathrm{eV} \rightarrow \mathrm{H}^{\oplus \oplus}{ }_{(\mathrm{g})}+e^{-}$$

We know $1 \mathrm{eV}=96.3 \mathrm{~kJ} \mathrm{~mol}^{-1}$

$$13.60 \mathrm{eV}=96.3 \times 13.60=1309.68 \mathrm{~kJ}$$

Thus, energy per $\mathrm{mol}=1309.68 \mathrm{~kJ}$

Q.27 Answer:

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Solution:

The electron gain enthalpy in kJ **is represented as**

$$\mathrm{Cl}_{(\mathrm{g})}+e^{-} \longrightarrow \mathrm{Cl}_{(\mathrm{g})}^{\ominus}+3.7 \mathrm{eV}$$

Now, $1 \mathrm{eV}$ **atom** $^{-1}=96.3 \mathrm{~kJ} \mathrm{~mol}^{-1}$

$$\therefore \quad 3.7 \mathrm{eV}=3.7 \mathrm{eV} \times 96.3 \mathrm{~kJ} \mathrm{~mol}^{-1}$$

Energy released when $35.5 \mathrm{~g}(1$ **mole) of chlorine is completely converted to** $\mathrm{Cl}^{\ominus}{ }_{(\mathrm{g})}$ **ion** $=3.7$, $96.3 \mathrm{~kJ}$

Energy released when $1 \mathrm{~g}$ **of chlorine is completely converted to** $\mathrm{Cl}^{\ominus}{ }_{(\mathrm{g})}$ **ion** $=\frac{3.7 \times 96.3}{35.5}=$ $10.04 \mathrm{~kJ}$

Q.28 Answer:

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Solution:

$$\left(\chi_{C}-\chi_{H}\right)=0.208\left[E_{A-B}-\sqrt{E_{A-A} \times E_{B-B}}\right]^{\frac{1}{2}}$$

$$=0.208 \times\left(\Delta_{C-H}\right)^{\frac{1}{2}}$$

$$=0.208 \times(6.0)^{\frac{1}{2}}=0.208 \times 2.5=0.52$$

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$$\chi_C - \chi_H = 0.52$$
$$\therefore \quad x_C = 0.52 + 2.1 = 2.62$$

Q.29 Answer:

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Solution:

The first inert gas which contains d-electron is in 4 th period (or from $3d$ transition element series), i.e.

$\mathrm{Kr}(Z = 36).$

Electronic configuration of Kr :

$1s^2, 2s^2 2p^6, 3s^2 3p^6 3d^{10}, 4s^2 4p^6$

Total number of d-electrons $= 10$

Total number of p-electrons $= 6 + 6 + 6 = 18$

Total number of s-electrons $= 2 + 2 + 2 + 2 = 8$

Difference in total number of $p-$ and $s-$ electrons $= (18 - 8)$

$= 10$

Atomic number of the inert gas is 36.

Q.30 Answer:

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Solution:

$\chi_{\mathrm{Cl}} = \frac{(\mathrm{IP})_{\mathrm{Cl}} + (\mathrm{EA})_{\mathrm{Cl}}}{5.6} = \frac{13+4}{5.6} = 3.03$

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